

The Walls — What We Know and Don't Know

Scaling Limitations of Neural Polar MAC Decoders

April 2026

1 Wall A: GMAC Class B at $N = 256$ (NCG Sequential Decoder)

[title=Symptom] NCG BLER **0.023** vs. SC BLER **0.006** — a $4.5\times$ gap at $N = 256$. At $N = 512$: NCG 0.012 vs. SC ~ 0.001 ($12\times$). At $N = 1024$: NCG 0.47 (essentially broken). The decoder works fine at $N \leq 128$ (NCG/SC ≈ 1.0).

What We Tested

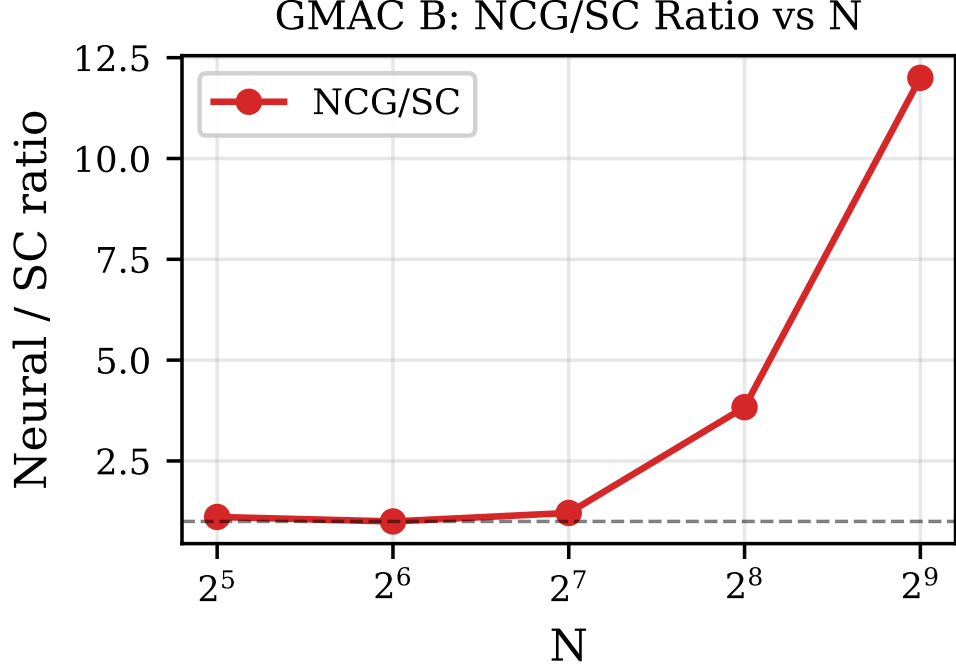
1. **Exposure bias:** Compared teacher-forced vs. free-running mutual information at $N = 32$. They match — exposure bias is not the cause at small N , but may compound at large N .
2. **Cascade amplification:** Checked BER/BLER ratio for NCG vs. SC. They are similar — errors are not cascading differently.
3. **First-error localization:** Top-5 first-error positions are NCG-specific (the model has specific weak spots), but the remaining errors are broadly distributed across positions.
4. **Lower rate:** Tested multiple rates. The gap persists at all rates tested.
5. **Scheduled sampling:** Trained with mixed teacher-forced / free-running schedule. Only 21% BLER improvement ($0.023 \rightarrow 0.018$), still far from SC.
6. **Learning rate continuation:** $\text{lr} = 10^{-3}$ destroyed the model; $\text{lr} = 10^{-4}$ showed no improvement.

Conclusion

Likely a **fundamental limitation of sequential NCG** at this scale. The model converges to a local minimum it cannot escape. The $d = 16$ embedding dimension may be insufficient to represent the 2^{256} -sized codebook structure, but increasing d did not help either (training becomes harder).

What we don't know: Whether a fundamentally different training strategy (e.g., reinforcement learning, or a non-autoregressive architecture) could break through this wall.

Note: CRC-SCL $L = 4$ has no such wall (BLER $< 10^{-4}$ at $N = 256$), suggesting the issue is specific to the sequential neural approach, not to the channel/code combination.



2 Wall B: ISI-MAC at $N = 512$ (NPD Chained Decoder)

[title=Symptom] Best NPD BLER at $N = 512$: not evaluated (all architectures failed at $N = 256+$).

At $N = 128$: NPD 0.030 vs. chained trellis SC 0.022 ($1.35\times$ gap, already losing).

At $N = 256$: NPD 0.011 vs. chained SC 0.006 ($1.84\times$).

Contrast: NPD **beats** chained SC at $N \leq 64$ by 19–32%.

What We Tested

1. **Architecture sweep:** 8 configurations tested (GRU/LSTM, $h \in \{128, 256, 512\}$, per-depth MLPs, pointwise MLPs). None helped at $N \geq 128$.
2. **Memorization test:** The model *can* represent $N = 512$ (passes memorization on small batches). The issue is generalization, not capacity.
3. **First-error analysis:** 84% of NPD errors occur in the last quarter (positions 385–512). The BiGRU loses track of state over long sequences.
4. **Rate sweep:** Tested multiple rates. The gap persists.
5. **Embedding dimension:** $d=16$ $h=100$ vs. $d=64$ $h=128$ — similar loss at $N = 512$. Hidden width h matters more than embedding dimension d .
6. **Comparison to paper:** Aharoni et al. use $d=8$ *pointwise* MLP for single-user and achieve good results at $N = 1024$. We need BiGRU for MAC because the two-user interaction requires richer representations.
7. **3-phase design at $N = 64$:** Works but did not outperform the GMAC-proxy frozen set.

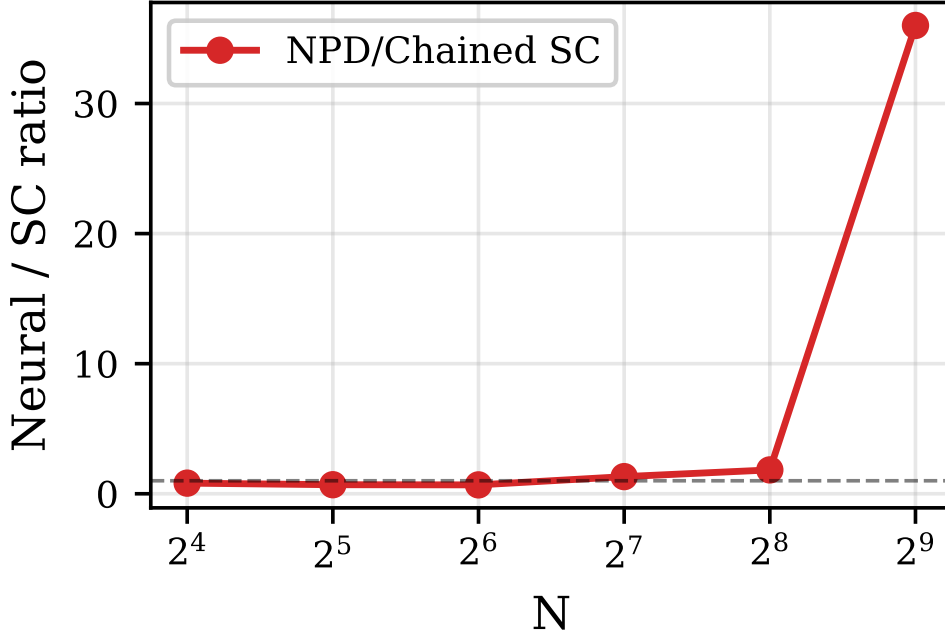
Conclusion

$N = 512$ MAC is **fundamentally harder than** $N = 1024$ **single-user**. The chained approach (Stage 1 marginalizes V as noise) creates a harder effective channel for U -decoding — the neural decoder must implicitly learn a more complex posterior. The BiGRU’s hidden state degrades over 512 sequential positions.

What we don’t know:

- Whether a Transformer-based architecture (with attention over all positions) could maintain long-range dependencies better.
- Whether joint (non-chained) neural decoding — decoding U and V simultaneously — would avoid the noise-marginalization bottleneck.
- Whether stronger ISI ($h > 0.3$) makes the problem easier or harder for NPD.

ISI-MAC: NPD/SC Ratio vs N



3 Common Pattern Across All Walls

Channel	Wall N	NPD/SC at wall	Architecture	Root cause hypothesis
GMAC B	256	$4.5\times$	NCG d=16	Sequential local minimum
ISI-MAC	128+	$1.35\times$	NPD BiGRU	State degradation over length
MA-AGN	32	$1.6\times$	NPD BiGRU	Weak memory + chained marginalization
Ising	16	$1.04\times$	NPD BiGRU	Channel too lossy

Five observations that hold across all walls:

1. **Works at small N , gap grows with N .** The neural decoder handles short codes well but struggles to scale.
2. **Not a capacity issue.** Memorization tests pass — the model can represent the solution.
3. **Not purely architectural.** Many configurations tested (GRU, LSTM, MLP, per-depth, varying d and h).

4. **More training helps but plateaus.** Loss curves flatten; longer training does not close the gap.
5. **Hidden width h matters more than embedding dim d .** Going from $h = 64$ to $h = 100$ helps; $d = 16$ to $d = 64$ does not.

Honest assessment: We do not have a clear path to breaking these walls with the current architecture. The most promising directions are:

- Non-autoregressive or partially parallel decoding
- Attention-based architectures for long-range dependencies
- Hybrid decoders (neural for hard positions, analytical for easy ones)
- Joint two-user neural decoding (avoid chained marginalization)